

Vellacombe Discovery Ltd

VD-18-131 | Route scout and delivery | Revision A

Final issue

Date: 29 Jan 2019

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1 Programme conclusion

Rennelwick Institute Drug Discovery Unit commissioned Route scout and delivery under VD-18-131, accession VA20181184. Client-supplied X-ray co-crystal structures guided selection of a less hindered substituent. Material was delivered for binding studies.

Table 1. Final programme disposition

Evidence	Disposition
Co-crystal structures	Provided by the client; crystallography performed by its collaborators
Chemistry direction	Less hindered substituent
Isolated material	Delivered for binding studies
Proposed structures	Design options, not delivered identities

2 Mandate

The quotation was dated 28 Sep 2018; preparation started on 05 Nov 2018. The programme closed on 28 Jan 2019 and this final issue is dated 29 Jan 2019.

The objective was to provide material that could test the structural hypothesis. A favourable fit in a structural model justified synthesis; it was not a binding measurement.

3 Responsibilities

Ruth Calderwood directed the analogue choice and route gates. Hassan Darwish led synthetic work and Clara Fenwick held analytical-release responsibility.

The client retained the structural biology and binding interpretation. Chemical identity, availability of material and receptor behaviour are separate conclusions.

4 Evidence boundary

The supplied programme export contains no compound-level result table. Individual yields, purity values, masses and identity spectra cannot be reproduced from that source.

The completed chemistry outcome is reported qualitatively. Original compound cards and delivery-lot documentation remain the sources for numerical particulars; no analytical value is reconstructed from a design drawing.

5 Structural hypothesis

The client-supplied X-ray co-crystal structures indicated that substituent crowding could be reduced. The chemistry direction was therefore a less hindered substituent.

This is a local design hypothesis. It does not establish that every other modification will preserve the pose or that the chosen material will improve binding.

6 Analogue definition

A proposed structure records what the team intends to make. A delivered identity requires isolated material linked to a compound card.

The distinction prevents an uncompleted preparation attempt from entering the binding-study inventory. Keep design drawings separate from the list of physically available material.

7 Route risk

The route gate concerns isolation of the intended analogue rather than disappearance of starting material. Related intermediates and side-products can make reaction conversion misleading.

Hassan Darwish's synthetic interpretation must therefore join the isolation history to the analytical assignment before a preparation is described as deliverable.

8 Route scout

The route scout documents the chemistry decision supporting the January readout. It is not evidence that recovery will be unchanged on another preparation scale.

For follow-up material, retain the intended structure and compare the isolated product with its earlier compound card. A changed route should receive its own isolation and identity assessment.

9 Isolated-yield basis

An isolated yield belongs to a defined preparation and material form. Recovery from a delivery vial cannot reconstruct that yield without the original starting-material basis.

No isolated-yield values are supplied in this archive export. The narrative delivery outcome is retained without inserting an assumed yield.

10 Identity handover

The compound-card identity must connect the label, the proposed target structure and the analytical assignment of the material actually isolated.

Clara Fenwick's analytical-release responsibility supports chemical identity. It does not establish receptor binding or validate the client's structural interpretation.

11 Purity context

Chromatographic purity is expressed as area percent with its method context. It is not a gravimetric composition and should not be converted directly into an assay concentration.

The receiving chemist should use the original compound card for purity and isolated-form particulars. Those particulars are not present in the numerical export supplied for this report.

12 Delivery-lot reconciliation

The programme outcome records delivery of material for binding studies. The delivery-lot label and compound card are the links between preparation and use.

Andrew May's sample-management function supports dispatch reconciliation. The client should reconcile received labels with the accompanying handover before preparing binding-study solutions.

13 Binding-study use

The delivered material allows the client to test whether reducing substituent crowding retains useful receptor binding. This report supplies the chemistry conclusion, not the biological answer.

Binding results should be linked to the actual compound-card identity and isolated form used. Comparisons against earlier material need the same traceability.

14 Readout link

RI13_route_readout_24Jan19.pptx is the related chemistry readout, archived with a 28 Jan 2019 file date. It describes the same programme and client-supplied structural guidance.

The date in that filename is retained as an archive identifier. This report uses 28 Jan 2019 as the programme completion date and 29 Jan 2019 as its issue date.

15 Inventory limitation

The supplied export records 0 in the compound-count field while its outcome records delivered material. That field is not treated as the physical delivery inventory.

The original handover is required to reconcile individual material labels. Neither an absent result table nor an administrative count supports inventing a compound list.

16 Next analogue

The next preparation should be chosen after the client reviews binding results and the X-ray co-crystal hypothesis. Further reduction in steric demand remains a proposal until that evidence is available.

Ruth Calderwood should define the comparison before changing several substituents together. Otherwise an unsuccessful binding result would be difficult to assign to the original crowding hypothesis.

17 Closure

The recorded turnaround is 84 calendar days and the programme price is GBP 39,300. These administrative values do not substitute for compound-level evidence.

The commissioned chemistry work is complete. Additional synthesis or binding work requires a new decision rather than being implied by this final report.

18 Final statement

A less hindered substituent was selected using client-supplied X-ray co-crystal structures, and material was delivered to Rennelwick Institute Drug Discovery Unit for binding studies. Crystallography was performed by the client's collaborators.

Isolated material is distinguished from proposed structures throughout this issue. Quantitative identity and recovery particulars remain with the original compound cards.

Name	Role	Signature	Date
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